Automated Landslide Risk Monitoring and Early Warning System Through AI for the North Eastern Region of India: A Case Study of Meghalaya

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Abstract : **Landslides are among the most destructive hazards in the North Eastern Region (NER) of India, and hazard management is still mainly based on static susceptibility maps that indicate where slopes are likely to fail, but not when they might fail. This paper presents an AI-based early warning system that combines a Random Forest (RF) susceptibility model (where) and a rainfall-threshold model (when), together with a low-cost LoRaWAN-based sensing and communication architecture adapted to the challenging terrain and limited connectivity of the region. Using Meghalaya as a case study, we clean the NASA Global Landslide Catalog to obtain 35 dated Meghalaya events (2007–2017). Of these, 89% are triggered by rainfall and 83% occurred in the June–September period, but only five are precisely located in East Khasi Hills. A first RF classifier trained on elevation, slope, aspect and land cover achieves a ROC-AUC of 0.73 on a test set with only four landslides. The AUC confidence interval is broad and overlaps the no-skill baseline, and the estimate is biased toward overestimation. The rainfall-threshold component has been scripted but not yet validated. Data limitations, deployment challenges and a phased development plan are discussed.**

Index Terms: landslide early warning, Random Forest, rainfall threshold, LSTM, IoT, LoRaWAN, North East India, Meghalaya

I. INTRODUCTION

Landslides kill people, damage infrastructure and isolate communities, especially in hilly areas. The North Eastern Region of India (NER) combines extremely high rainfall with young, structurally weak geology, steep slopes and high seismicity (most of the region lies in Seismic Zone V), and a rapidly increasing human presence in the form of roads, mining, forest clearance and shifting (jhum) cultivation. Meghalaya paints a clear picture: the Sohra (Cherrapunji) and Mawsynram areas commonly report annual rainfall of 10,000–12,000 mm, most of it during the monsoon months (May–September).

Operational warning in NER is scarce. GIS susceptibility maps show where landslides are likely, and national guidance recommends landslide early warning [1], [2], yet these maps do not account for antecedent rainfall, soil saturation or slope movement. Elsewhere, rainfall intensity–duration thresholds [3], [4], [5] are used operationally, but they require long rainfall and landslide records that are uncommon in NER.

This work makes four contributions. We (i) introduce a hybrid approach combining a Random Forest-based susceptibility model with a rainfall-threshold model, (ii) provide a reproducible pipeline for transforming the NASA Global Landslide Catalog (GLC) [6], [7] into a validated Meghalaya inventory, (iii) train an initial susceptibility model and discuss its statistical uncertainty and sampling limitations openly, and (iv) propose an inexpensive sensing and communication architecture and enumerate the challenges of deploying it in NER. The rainfall-threshold component has not been validated, and we say so where relevant.

II. RELATED WORK

A. Global and Indian Warning Systems

Landslide warning systems in Hong Kong, Italy and Japan use rainfall thresholds based on long historical records, converted into easily understood warning levels. Intensity–duration thresholds originate with Caine [3] and have been compiled and reviewed worldwide [4], [5]. In India, the nodal agency for hazard mapping is the Geological Survey of India (GSI), and the National Remote Sensing Centre provides an inventory and susceptibility map through the Bhuvan geoportal. A comprehensive review of rainfall-induced landslide studies in the Indian Himalaya [8] charts the extent of susceptibility and threshold modelling work, though published evidence on operational real-time warning in NER remains sparse.

B. Machine Learning and Monitoring Technologies

Logistic regression is a common baseline. Ensemble models such as Random Forest [9] are favored for susceptibility mapping for their tolerance to noisy data and ability to rank features. Support vector machines and neural networks are also used, and convolutional networks have been applied to imagery for landslide detection. LSTM networks [10] suit rainfall time series because they learn the antecedent effects that drive triggering. Comparative studies generally favor ensemble models for static classification and sequence models for time-dependent triggering, which explains the hybrid architecture of this work. On the monitoring side, ground instrumentation (rain gauges, soil moisture and pore-pressure sensors, tilt meters, GNSS) is complemented by satellite InSAR for slow deformation and by LPWAN technologies (e.g., LoRaWAN) for connectivity in remote terrain. We found little work that combines AI-based dynamic

prediction with a monitoring architecture designed for the terrain and connectivity of NER, which we address here.

III. STUDY AREA AND DATA

A. Meghalaya

Meghalaya lies on an uplifted, table-topped plateau of ancient crystalline and sedimentary rock (the Khasi, Jaintia and Garo Hills), dissected by steep valleys and gorges, with elevation from near sea level to well over 1,900 m. Monsoon winds from the Bay of Bengal are forced up the southern escarpment and produce some of the highest recorded rainfall in the world. Weak, heavily fractured sandstone, shale and coal-bearing rocks, overlain by residual soils and cut by roads, shallow coal and mineral mines, jhum cultivation and construction, are prone to shallow translational slides, debris flows and rockfalls. East Khasi Hills, which contains Shillong and Sohra, is the main study area.

B. Landslide Inventory

We used the NASA GLC [6], [7], [11], a catalogue of reported rainfall-related landslides, mostly from the news, containing 11,033 global entries with 31 fields. We reorganised and screened it in a Python pipeline (pandas and GeoPandas, summarised in Table I). GADM [12] district polygons were used to attach each record to a state and district independently of the catalogue text; this also recovered 235 Indian records that lacked a country label. This gives 1,497 Indian records, 462 in the eight NER states and 36 within Meghalaya. One of them (event 141, "Singtam", a town in Sikkim) appears to be mis-located, and removing it leaves 35 events on 26 different days, 20 of which state a positional accuracy of 5 km or better. The time of day is known for 16. The events are plotted in Fig. 1.

TABLE I
CLEANING OF THE NASA GLC

Step	Operation	Records
1	Load and parse dates	11,033
2	Eliminate duplicates (date, coordinates, title)	11,004
3	India filter (country label + polygon)	1,497
4	North Eastern Region states	462
5	Inside Meghalaya polygon	36
6	Exclude mis-geocoded record	35
7	East Khasi Hills, accuracy ≤ 5 km	5

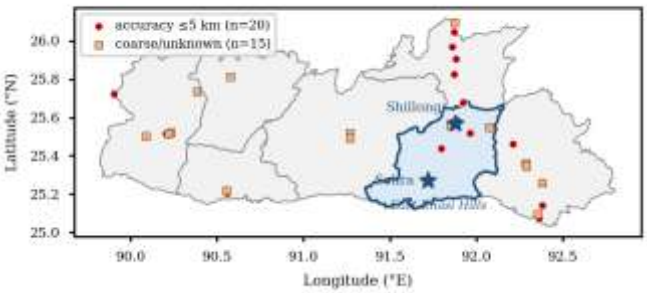


Fig. 1. GLC landslides in Meghalaya, 2007–2017 (n = 35). East Khasi Hills is outlined; districts are from GADM (pre-2012).

C. Terrain and Land Cover

Elevation, slope and aspect were calculated from a 30 m Copernicus DEM [13], and land cover from ESA WorldCover 2021 at 10 m [14].

IV. METHODOLOGY

A. Hybrid Framework

The framework separates *where* from *when*. Spatial susceptibility is derived from terrain and land cover with a static Random Forest. The probability of a landslide being triggered by current and antecedent rainfall is derived from a dynamic rainfall-threshold model; the two outputs are combined and reclassified into a risk class (Green, Yellow, Orange or Red).

B. Spatial Susceptibility Model

Landslide samples are the 20 GLC events with positional accuracy of 5 km or better (17 rainfall-triggered, two of unknown trigger and one quarry failure). Background samples are 100 random points representing non-landslide conditions. The predictors are elevation, slope, aspect and land-cover class at each point. The classifier is a Random Forest with 100 trees and balanced class weights, evaluated on a held-out test set of 24 samples (20 non-landslide and 4 landslide). Lithology, soil, curvature, distance to roads and drainage, and NDVI are proposed as extensions.

C. Rainfall-Threshold Model

For each Meghalaya event we extract a 60-day window of daily precipitation from GPM IMERG Final Run (V07, 0.1°) [15] at the nearest grid cell. From it we derive 1-, 3-, 5-, 7-, 10-, 15- and 30-day accumulations and the maximum daily intensity, for the event day and for non-event days at least three days from any catalogued event. Planned analyses are empirical lower-bound (10th percentile) thresholds, an intensity–duration curve, and logistic regression and Random Forest baselines under cross-validation grouped by event. An LSTM is proposed once the inventory is large enough. The extraction and analysis scripts exist but have not yet been run.

D. Evaluation

Missed events are costlier than false alarms, so we report precision, recall, F1-score and ROC-AUC, and we give

confidence intervals for the AUC following Hanley and McNeil [16].

V. PROPOSED SYSTEM ARCHITECTURE

The architecture has four layers (Fig. 2). The sensing layer comprises tipping-bucket rain gauges, soil moisture probes, MEMS tilt meters, piezometers, optional low-cost GNSS receivers and satellite feeds (Sentinel-1 InSAR and Sentinel-2). The edge layer uses solar-powered LoRaWAN nodes that report to gateways on towers or forest outposts, with filtering and compression at the gateway and cellular or satellite backhaul. The cloud layer hosts the models, a rules engine that fuses the spatial class with the rainfall-threshold probability into a risk class, and a database for retraining. The alert layer sends SMS messages, feeds dashboards for the State Disaster Management Authority and district administration, links to sirens and public-address systems, and issues multilingual alerts (Khasi, Garo, English and Assamese). This architecture is a design contribution; no sensor network has been deployed.

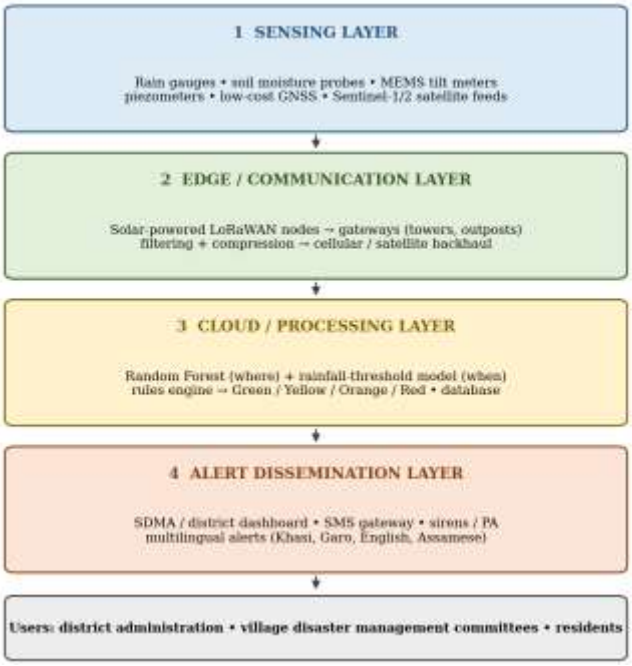


Fig. 2. Proposed four-layer early warning architecture.

VI. RESULTS AND DISCUSSION

A. Inventory Analysis

Of the 35 Meghalaya events, 31 (89%) are rainfall-triggered, three have an unknown trigger and one is a quarry failure; 29 (83%) occur in June–September (Fig. 3). Reported fatalities total 48 (counts are given for 31 events). Annual counts peak in 2015 (11 events), but this reflects media coverage and should not be read as a trend. Seven events plot inside East Khasi Hills, five of them with positional accuracy of 5 km or better (Shillong 2010, 2014 and 2015; Lamlyer 2017; Mawjrong 2017). No record lies within 20 km of Sohra or Mawsynram, which reflects the media-based nature of the catalogue and not an absence of landslides.

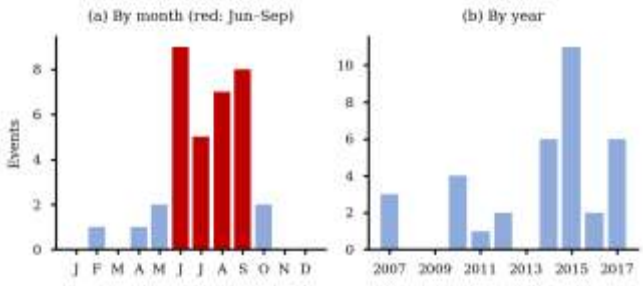


Fig. 3. Distribution of the 35 Meghalaya events by month and by year.

B. Random Forest Performance

Table II reports the classification results. Overall accuracy was 75% and the ROC-AUC was 0.73 (Fig. 4). For the landslide class, precision, recall and F1-score were all 0.25: the model found one of the four test landslides, and one of its four landslide predictions was correct. Accuracy is dominated by the 20 non-landslide samples, and a classifier that always predicted "no landslide" would score 83% on this test set.

TABLE II
RANDOM FOREST RESULTS ON THE TEST SET

Class	Precision	Recall	F1-score	Support
No landslide	0.85	0.85	0.85	20
Landslide	0.25	0.25	0.25	4
Accuracy			0.75	24
Macro avg.	0.55	0.55	0.55	24
Weighted avg.	0.75	0.75	0.75	24

ROC-AUC = 0.7313

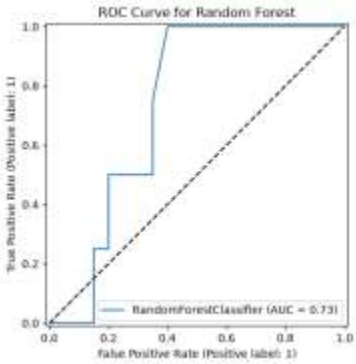


Fig. 4. ROC curve of the Random Forest on the test set (AUC = 0.73; 4 landslide and 20 non-landslide samples).

C. Statistical Uncertainty

With four landslides in the test set, a Hanley–McNeil 95% confidence interval for an AUC of 0.73 runs from about 0.43 to 1.00, which includes performance no better than chance. We repeated the experiment on the same 20 events and 100 background points drawn inside Meghalaya. Repeated stratified 5-fold cross-validation (50 repeats) gave a mean AUC of 0.66 (95% interval 0.53–0.73), and 300 random 80/20 splits gave a median of 0.68 (5th–95th percentile 0.46–0.89). The reported 0.73 therefore lies within the range that chance variation in a single split can produce.

D. Sampling Weaknesses

Inspection of the training data revealed two problems. First, the background points were drawn over the rectangular extent of the DEM and land-cover rasters, about 61% of which lies outside Meghalaya. Of the background points, 56% lie below 100 m elevation, against about 13% of random points inside the state, and 10 of 100 have no land-cover data. A model trained this way can score well by separating the Meghalaya plateau from the surrounding plains rather than by learning slope-failure susceptibility. Under the same cross-validation procedure, this design gave an AUC of about 0.81, against about 0.64 when the background was restricted to the state. Second, slope was computed from raw per-pixel gradients instead of metric units, so the values are not in degrees (landslide points average 73° against about 10° when computed correctly). Tree models depend only on rank order (Spearman $\rho = 0.99$ between the two versions), so the fit is barely affected, but the model would mislead if applied to slopes in degrees. The reported AUC of 0.73 should therefore be regarded as optimistic.

E. Feature Importance

Elevation (0.37) and slope (0.31) dominate, followed by aspect (0.23) and land cover (0.09) (Fig. 5). Given the sampling issue above, and because catalogue events cluster near roads and settlements (landslide points average 706 m elevation against 591 m for random in-state points, and sit on gentler 30 m slopes, about 11° against 15°), these rankings should not be read as physical susceptibility.

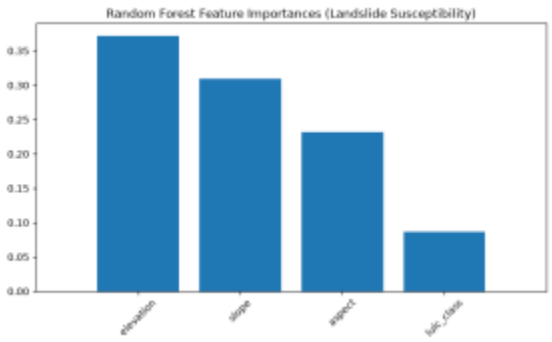


Fig. 5. Random Forest feature importances.

F. Pending Results

The rainfall-threshold results, a susceptibility zonation map and a combined-risk demonstration (for example the 17 June 2017 Mawjrong landslide in East Khasi Hills) depend on the GPM extraction, which has not been run. With about 30 rainfall-triggered events, an LSTM can be trained only with substantial data augmentation, so simpler baselines will be reported alongside it.

VII. CHALLENGES AND LIMITATIONS

Data. The inventory is small, media-based and ends in 2017; only 20 of 35 events have positional accuracy of 5 km or better, and 15 carry uncertainty of up to 50 km. *Terrain and access.* Rugged, forested terrain makes sensor installation and maintenance costly, and many high-risk slopes lie far from roads. *Connectivity and power.* Cellular coverage is patchy and grid power unreliable, which motivates LoRaWAN and solar nodes, although heavy monsoon cloud can limit solar charging. *Institutions and community.* Long-term funding and coordination across GSI, IMD, the State Disaster Management Authority and local bodies are hard to sustain, and alerts must be trusted and understood across several languages. *Seismicity.* Earthquakes can alter slope conditions in ways that rainfall-threshold models do not capture.

In addition, the Random Forest used only four predictors and was tested on 24 samples; the framework has not been validated in the field; and no rainfall–landslide relationship has yet been quantified.

VIII. CONCLUSION AND FUTURE WORK

We proposed an AI-based early warning framework for NER and examined it on Meghalaya. The cleaned catalogue confirms the monsoon control on landslides (89% rainfall-triggered, 83% in June–September), but it also shows that the available inventory is small. The initial Random Forest achieved an AUC of 0.73 on only four test landslides, which is far too uncertain, and probably optimistic, to be operationally viable. Next steps include: (1) running the GPM extraction and testing thresholds and baselines; (2) retraining the susceptibility model (background points only within the state, metric slope, circular encoding of aspect, more layers and spatially blocked cross-validation); (3) increasing the inventory with GSI, SDMA and news-archive records and field campaigns; (4) piloting a sensor network along the Shillong–Sohra road corridor; and (5) extending to other NER states, incorporating InSAR deformation data and collaborating with disaster management authorities.

ACKNOWLEDGMENT

The authors are grateful for the freely available data from NASA, the European Space Agency and the Copernicus programme that were used in this paper.

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